

Git & GitHub Workflow Guide

Branching, commits, pull requests and recovery commands for small teams.

Branching model

- main is always deployable — protect it and merge through pull requests.
- Create short-lived feature branches: feature/<ticket>-<slug>.
- Rebase your branch on main before opening a PR to keep history linear.
- Delete branches after merge; keep the remote tidy.

```
git switch -c feature/FIL-12-search-filter
git fetch origin && git rebase origin/main
git push -u origin HEAD
```

Everyday commands

Goal	Command
Status / diff	git status · git diff --staged
Stage part of a file	git add -p
Amend the last commit	git commit --amend --no-edit
Undo last commit (keep changes)	git reset --soft HEAD~1
Discard local changes	git restore <file>
Find the commit that broke it	git bisect start / good / bad
See who last touched a line	git blame -L 40,60 <file>
Stash work in progress	git stash push -m "wip" · git stash pop

Conventional commits

A consistent prefix makes changelogs and semantic releases mechanically generatable.

```
feat(library): add category filter chips
fix(admin): guard against duplicate filenames
docs(readme): document the export workflow
chore(deps): bump vite to 8.x
Types: feat · fix · docs · style · refactor · perf · test · chore
```

Pull-request checklist

- Title states the outcome, description explains why and how it was verified.
- Keep the diff reviewable — under ~400 changed lines where possible.
- Add screenshots for UI changes and note any new configuration or environment variables.
- CI green, no merge conflicts, at least one reviewer approval.

Resolving conflicts

```
git merge origin/main      # or: git rebase origin/main
# edit the conflicted files, then:
git add <files>
git rebase --continue      # or: git commit
```

Recovery

- git reflog shows every position HEAD has held — almost nothing is truly lost.
- git reset --hard <sha> moves the branch; git revert <sha> adds an inverse commit (safe on shared branches).
- git checkout <sha> -- <file> restores a single file from history.
- git clean -nd previews untracked files; -fd deletes them.

Good .gitignore hygiene

```
node_modules/
dist/
.env
.env.local
*.log
.DS_Store
```